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I am a student of BS in Data Science from Indian Institute of Technology currently pursuing Diploma level. I like to learn new technologies and make valuable things out of them that makes life easy.

AI/LLM usage

Around 20% limited to code suggestion and documentation formatting along with UI tweaks

Description

This project demanded the development of a Placement Portal Application that can be used by Admins, Company recruiters and students in order to get easy placement drive experience.

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Technologies used

Flask Backend web framework used to build the REST API.

Flask-RESTful Simplifies creation of RESTful API resources and endpoints.

SQLAlchemy ORM used for database modeling and CRUD operations.

SQLite Lightweight relational database for storing application data.

Vue.js Frontend framework for building a responsive single-page application.

Bootstrap 5 CSS framework for responsive UI components and styling.

Flask-JWT-Extended Implements JWT-based authentication and authorization.

Flask-Bcrypt Securely hashes user passwords before storing them.

Redis Caching layer and message broker for asynchronous tasks.

Celery Executes background jobs such as scheduled reports and reminders.

Vue Router Handles client-side routing and navigation.

Axios Performs HTTP requests between the Vue frontend and Flask backend.

Python Primary programming language for backend development.

JavaScript Used to implement frontend logic in Vue.js.

HTML5 Provides the structure of the web pages.

CSS3 Used for custom styling alongside Bootstrap.

DB Schema Design



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API Design

Method	Endpoint	Description
POST	/register	Register a new user
POST	/login	Authenticate user and issue JWT
GET	/student/profile	Retrieve student profile
PUT	/student/profile	Update student profile
GET	/company/profile	Retrieve company profile
POST	/drives	Create a placement drive
GET	/drives	View available placement drives

POST	/applications	Apply for a placement drive
GET	/applications	View application status
GET	/branches	Retrieve available branches

Architecture and Features

The Placement Portal follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The frontend is developed using Vue.js and Bootstrap, providing a responsive user interface that communicates with the backend through RESTful APIs. The backend is built using Flask and is organized into modular components. The `controllers` directory contains API resources and business logic for authentication, student, company, admin, placement drive, branch, and application management. The `models.py` file defines the database schema and relationships using SQLAlchemy ORM, while the database configuration is maintained separately. Authentication is implemented using JWT, ensuring secure access to protected endpoints. SQLite serves as the primary database, with SQLAlchemy handling database operations. Redis and Celery are integrated for caching and executing asynchronous background tasks. This modular architecture improves code maintainability, scalability, and separation of concerns.

The system supports three user roles: Administrator, Student, and Company, each with dedicated functionalities and role-based access control. Students can register, complete their profiles, browse approved placement drives, apply for eligible drives, and track their application status. Companies can create and manage company profiles, submit placement drives for approval, and review student applications. Administrators can manage branches, approve or reject company registrations and placement drives, monitor users, and oversee the entire placement process through an administrative dashboard. The application implements secure JWT-based authentication, password hashing using Bcrypt, profile completion tracking, eligibility-based filtering of placement drives, unique application validation to prevent duplicate submissions, and comprehensive CRUD

operations for all major entities. Additional features include branch management for eligibility mapping, many-to-many relationships between drives and branches, background task support through Celery, Redis caching, and a responsive user interface built with Vue.js and Bootstrap. Together, these features provide a secure, scalable, and efficient placement management system.

Video

 [MAD2_placement_portal](#)